COSC3106: Theory of Computing

CH#4:

Lect#10: Pumping Lemma and Nonregular Languages

School of Computer Science and Technology

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This lecture

- In this lecture, you will …
  - Learn how to prove that certain languages are not regular.

- Is every language regular?
  - The answer is NO.
  - What language could be a candidate for not being regular?
  - Ex: $\{a^n b^n : n \geq 0\}$,

$$\Rightarrow \{\varepsilon, ab, aabb, aaabbb, aaaabbbb\}$$

    - There are two condition in this language:
      - $\#a = \#b$
      - All $a's$ are to the left of $b's$

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Idea Behind Pumping a String

- A DFA has finite states.
- If a string is long enough, the DFA must:
  - revisit a state
  - create a loop?
- This means part of the string can be repeated.

- Ex:

$$q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3 \ldots\ldots\ldots$$

  - That means loop exist → we can repeat part of the string.

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Pumping Lemma:

- Let A be a regular language. Then there exists an integer $p \geq 1$, called the pumping length, and the following holds:
  - Every string s in A, with $|s| \geq p$, # states $\leq p$, can be written as $s = xyz$, such that:
    - $|xy| \leq p$
    - $|y| > 0$, $(i.e., y \neq \varepsilon)$
    - For all $i \geq 0, xy^iz \in A$.

- $|xy| \leq p$: The loop happens within the first p characters.
- $|y| > 0$: y cannot be empty. Because the loop must contain something.
- $xy^iz \in A$: We can repeat y any number of times and still remain in the language.

- **Ex#1**: Let Σ be the alphabet of L. Since L is a regular language, there exists a DFA $M = (Q, \Sigma, \delta, q, F)$ that accepts L. Consider the language:

$$L = \{ a^n b^n : n \geq 0 \}$$

  - We will prove by contradiction that L is not a regular language.

Solution

- Step 1: Assume L is regular, so Pumping Lemma must hold.
- Step 2: We define p (pumping length) $\geq$ the number of states in Q, and choose a string $s \in L$:

$$s = a^p b^p$$

$$\Rightarrow |s| = 2p \geq p$$

$a\ a\ a\ a\ a\ a\ a$ ……	$b\ b\ b\ b\ b\ b\ b\ b$ ……

- Step 3: PL states that s can be written as xyz, and;

$$|\ xy\ | \leq p, \quad \text{both x and y contain only } a\text{'s.}$$

- Step 4: Pump
  - Take $i = 2 \Rightarrow s = xy^2z = xyyz$ is this true? No
    - Because of that:
      - $|y| \geq 1$ (non-empty)
      - It means that we added a

$$\Rightarrow \#a \neq \#b$$

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- **Ex#2**: Let Σ be the alphabet of L. Since L is a regular language, there exists a DFA $M = (Q, \Sigma, \delta, q, F)$ that accepts L. Consider the language:

$$L = \{\, ww : w \in \{a, b\}^* \,\}$$

  - We will prove by contradiction that L is not a regular language.

Solution

- Step 1: Assume the opposite L is regular, so Pumping Lemma must hold.
- Step 2: We define p (pumping length) $\geq$ the number of states in Q, and choose a string $s \in L$:

$$|s| = a^p b a^p b = 2p + 2 \geq ps$$

$a\ a\ a\ a\ a\ a\ a\ \ b$	$a\ \ a\ \ a\ \ a\ \ a\ \ a\ \ a\ \ b$

- Step 3: PL states that s can be written as xyz, and;
  - $|\, xy \,| \leq p$, both x and y contain only $a^p b$, or a^p only
  - $|y| \geq 1$: the substring y lies entirely within the first p symbols of s.
  - Those first p symbols are all a's, so y must consist only of a's.
- Step 4: Pump
  - Take $i = 2 \Rightarrow s = xy^2z = xyyz$ is this true? No
    - Because of that:
      - $|y| \geq 1$ (non-empty)
      - It means that we added a

$\Rightarrow \#a\ (1st\ half) \neq \#a\ (2nd\ half) \Rightarrow nonregular\ language$

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